

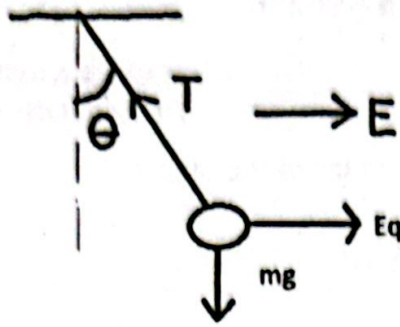
SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI - 600 063
PERIODIC TEST 1 (2026 - 2027)
PHYSICS - SET 1

CLASS: XII
DATE: 12.06.2026

MAX. MARKS: 20
DURATION: 40 MINS

- X 1. If an object is positively charged, theoretically the mass of the object ____
 - a. Increases slightly by a factor of 9.11×10^{-31} kg
 - b. Decreases slightly by a factor of 9.11×10^{-31} kg
 - c. Remains the same
 - d. May increase or decrease
- X 2. Two equally charged spheres attract each other. Now if they touch each other then they will ____
 - a. Attract each other with the same amount of force
 - b. Repel each other
 - c. Attract each other with less amount of force
 - d. not attract or repel
- ✓ 3. Three equal positive charges are kept at the corners of an equilateral triangle. What will be the vector sum of the forces acting on the particles?
 - a. Directed towards the center
 - b. Directed radially outside
 - c. Acts along one of the sides of the triangle
 - d. Zero
- X 4. 3 coulomb = ____ Electro Static Unit.
 - a. 9×10^9
 - b. 3×10^8
 - c. 1.602×10^{-19}
 - d. 2×10^9
- ✓ 5. Which among the following cannot be the charge of a charged body?
 - a. 4.8×10^{-19} Coulomb
 - b. 6.4×10^{-19} Coulomb
 - c. 5×10^{-19} Coulomb
 - d. 3.2×10^{-19} Coulomb
- ✓ 6. The amount of force exerted on a unit positive charge in an electric field is known as ____
 - a. Electric field intensity
 - b. Electric flux
 - c. Electric potential
 - d. Electric lines of force
- X 7. Two point charges of the same polarities are hung with the help of two threads and kept close. The angle between the threads will be ____ if the system is taken to space.
 - a. 180 degree
 - b. 90 degree
 - c. 45 degree
 - d. 60 degree
- ✓ 8. Two point charges $+4q$ and $+q$ are kept at a distance of 30 cm from each other. At which point between them, the field intensity will be equal to zero?
 - a. 15cm away from the $+4q$ charge
 - b. 20cm away from the $+4q$ charge
 - c. 7.5cm away from the $+q$ charge
 - d. 5cm away from the $+q$ charge
- ✓ 9. Electric flux will be maximum if the angle between the field lines and area vector is ____
 - a. 45°
 - b. 135°
 - c. 90°
 - d. 0°
- ✓ 10. Gauss's Law cannot be applied in ____
 - a. Hollow sphere
 - b. Solid sphere
 - c. Cube
 - d. Unbounded surface

11. A ball of 80mg mass and a 2×10^{-8} C charge is hung with a thread in a uniform horizontal electric field of 2×10^4 V/m. What is the angle made by the thread with vertical? (Given $\tan^{-1} 0.51 \approx 27^\circ$ and $\tan^{-1} 0.867 \approx 30^\circ$)



- a. 27° b. 30° c. 45° d. 0°

12. _____ is not the unit for electric field.

- a. N/C b. V/m
c. ampere ohm/m d. N/C²

13. When the reference point is along the axial line of a dipole, the electric field at that point is

- a. parallel to the dipole moment b. perpendicular to the dipole moment
c. inclined at 30° with the dipole moment d. None of the above

14. The electric field due to an infinite plane sheet of charge is:

- a. inversely proportional to distance b. directly proportional to distance
c. independent of distance d. zero

15. If dielectric constant increases, the electrostatic force between charges in the medium

- a. increases b. decreases c. remains same d. becomes zero

16. The torque on an electric dipole in uniform field is maximum when angle is:

- a. 0° b. 30° c. 90° d. 180°

17. If a charge is placed at the Centre of a cube, the flux through one face is:

- a. q/ϵ_0 b. $q/6\epsilon_0$ c. $q/4\epsilon_0$ d. zero

18. The area vector of a surface of area 0.01 m^2 kept in x-y plane enclosed with the flux of $1.6 \times 10^8 \text{ Nm}^2\text{C}^{-1}$ is

- a. $0.01 \hat{k} \text{ m}^2$ b. $1.6 \times 10^8 \text{ m}^2$
c. $3.2 \times 10^8 \text{ m}^2$ d. $0.016 \times 10^8 \hat{k} \text{ m}^2$

19. Electric field at distance r from a point charge is E. What will be the field at distance 2r?

- a. E/2 b. E/4 c. 2E d. 4E

20. Electric field due to infinite plane sheet is E. If charge density doubles, new field is

- a. E/2 b. E c. 2E d. 4E

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SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI - 600 063
TEST- 1 (2026 - 2027)
PHYSICS

CLASS: XII
DATE: 17.06.2026

MAX. MARKS: 35
DURATION: 1hr. 30 mins.

GENERAL INSTRUCTIONS:

1. Question no. 1-6 carry 1 mark each.
2. Question no. 7 and 8 are assertion and reasoning questions.
3. Question no. 9 and 10 carry 2 marks each.
4. Question no. 11, 12 and 13 carry 3 marks each.
5. Question no. 14 is case-based carrying 4 marks totally.
6. Question no. 15 and 16 are long answer questions carrying 5 marks each.
7. Answer all the questions and draw the diagram wherever necessary.

1. How many electrons must be removed from an electrically neutral metal plate to give it a positive charge of 1×10^{-7} coulomb?

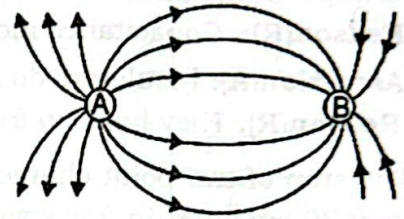
(a) 6.25×10^{11} (b) 6.45×10^{13} (c) 6.25×10^{-11} (d) 6.45×10^{-13}

2. The unit of permittivity of free space (ϵ_0) is

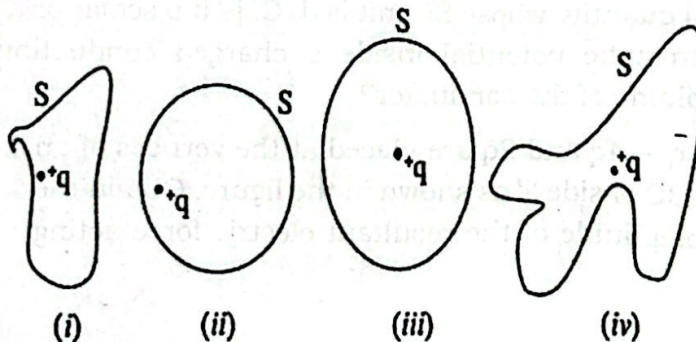
(a) $\text{CN}^{-1}\text{m}^{-1}$ (b) Nm^2C^{-2} (c) $\text{C}^2\text{N}^{-1}\text{m}^{-2}$ (d) $\text{C}^2\text{N}^{-2}\text{m}^{-2}$

3. The spatial distribution of the electric field due to two charges (A, B) is shown in figure. Which one of the following statements is correct?

- (a) A is +ve and B is -ve and $|A| > |B|$
(b) A is -ve and B is +ve, $|A| = |B|$
(c) Both are +ve but $A > B$
(d) Both are -ve but $A > B$

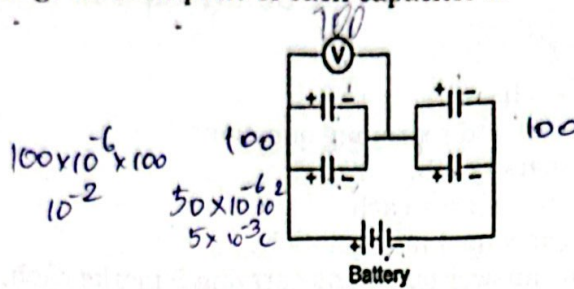


4. The Electric flux through the surface



- (a) in Fig. (iv) is the largest.
(b) in Fig. (iii) is the least.
(c) in Fig. (ii) is same as Fig. (iii) but is smaller than Fig. (iv)
(d) is the same for all the figures.

5. In bringing an electron towards another electron, the electrostatic potential energy of the system
 (a) increases (b) decreases (c) remains unchanged (d) becomes zero
6. Four capacitors, each of $50 \mu\text{F}$ are connected as shown. The DC voltmeter V reads 100 V . The charge on each plate of each capacitor is



- (a) $2 \times 10^{-3} \text{ C}$ (b) $5 \times 10^{-3} \text{ C}$ (c) 0.2 C (d) 0.5 C

READ THE FOLLOWING QUESTIONS AND CHOOSE:

- (a) If both Assertion(A) and Reason(R) are true and Reason(R) is correct explanation of Assertion(A)
 (b) If both Assertion(A) and Reason(R) are true but Reason(R) is not correct explanation of Assertion(A)
 (c) If Assertion(A) is true but Reason(R) is false
 (d) If Assertion(A) is false and Reason(R) is true.

7. **Assertion(A):** The plates of a parallel plate condenser are connected to a battery. Charge on the plates increases on introducing a dielectric slab between the plates.
Reason(R): Capacitance increases and hence more charge is drawn from the battery.

8. **Assertion(A):** Insulators do not allow flow of current through themselves.
Reason(R): They have no free charge carriers.

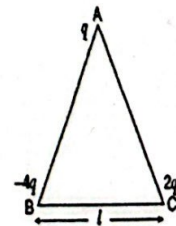
9. The sum of two-point charges is $7 \mu\text{C}$. They repel each other with a force of 1 N when kept 30 cm apart in free space. Calculate the value of each charge.

10. (a) Depict the equipotential surfaces for a system of two identical positive point charges placed at a distance of 'd' apart.

- b) Name the physical quantity whose SI unit is J/C . Is it a scalar or a vector quantity?

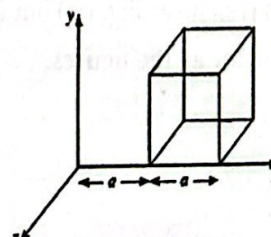
Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor?

11. Three-point charges q , $-4q$ and $2q$ are placed at the vertices of an equilateral triangle ABC of side 'l' as shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge q .



12. State Gauss's law in electrostatics. A cube with each side 'a' is kept in an electric field given by $\vec{E} = Cx\hat{i}$, (as is shown in the figure) where C is a positive dimensional constant. Find out

- (i) the electric flux through the cube, and
 (ii) the net charge inside the cube.



13. Two charges of values $2 \mu\text{C}$ and $-50 \mu\text{C}$ are placed 80 cm apart. Calculate the distance of the point from the smaller charge where the intensity is zero.

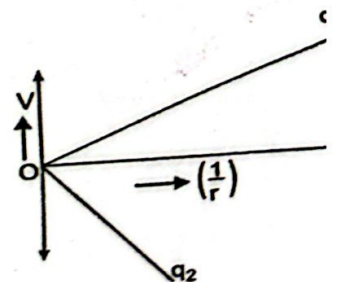
14. CASE BASED QUESTIONS

READ THE PARAGRAPH AND ANSWER THE QUESTIONS GIVEN CAPACITOR AND CAPACITANCE

A capacitor contains two oppositely charged metallic conductors at a finite separation. It is a device by which capacity of storing charge may be varied simply by changing separation and/or medium between the conductors. The capacitance of a capacitor is defined as the ratio of magnitude of charge (Q) on either plate and potential difference (V) across the plate, i.e.,

$$C = \frac{Q}{V}$$

- i. Three capacitors each of capacitance 9 pF are connected in series:
The total capacitance of the combination is
 a. 3 pF b. 5 pF c. 6 pF d. 9 pF
 - ii. What is the potential difference across each capacitor if the combination is connected to 120 V supply?
 a. 40 V b. 60 V c. 50 V d. 70 V
 - iii. Net capacitance of three identical capacitors in series is $1 \mu\text{F}$. What will be their net capacitance if connected in parallel?
 a. $6 \mu\text{F}$ b. $3 \mu\text{F}$ c. $9 \mu\text{F}$ d. $12 \mu\text{F}$
 - iv. Find the ratio of energy stored in the two configurations if they are both connected to the same source.
 a. $2:9$ b. $1:9$ c. $3:9$ d. $1:1$
5. a) Find expressions for the force and torque on an electric dipole kept in a uniform electric field.
 b) Two identical point charges, q each, are kept 2 m apart in air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q .
 c) What is the amount of work done in moving a point charge Q around a circular arc of radius ' r ' at the centre of which another point charge ' q ' is located?
16. a) Briefly explain the principle of a capacitor. Derive an expression for the capacitance of a parallel plate capacitor, when the space between the two plates are completely filled by a dielectric medium.
 b) The two graphs are drawn below, show the variations of electrostatic potential (V) with $1/r$ (r being the distance of field point from the point charge) for two point charges q_1 and q_2 .
- (i) What are the signs of the two charges?
 - (ii) Which of the two charges has the larger magnitude and why?



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TEST- 2 (2026 - 2027)
PHYSICS

ROLL NO. :

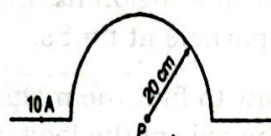
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CLASS: XII
DATE: 25.07.2026

MAX.MARKS: 35
DURATION: 1hr. 30 mins.

GENERAL INSTRUCTIONS:

1. Question no. 1-8 carry 1 mark each.
2. Question no. 7 and 8 are assertion and reasoning questions.
3. Question no. 9 and 10 carry 2 marks each.
4. Question no. 11, 12 and 13 carry 3 marks each.
5. Question no. 14 is case-based carrying 4 marks totally.
6. Question no. 15 and 16 are long answer questions carrying 5 marks each.
7. Answer all the questions and draw the diagram wherever necessary.

1. Two rods of copper and aluminium of each equal length 20 cm and equal cross-sectional area 2 mm^2 . If they are joined in series, find the resistance of the combination. Resistivity of copper = $1.7 \times 10^{-8} \Omega \text{ m}$ and resistivity of aluminium = $2.6 \times 10^{-8} \Omega \text{ m}$.
a) $4 \times 10^{-8} \Omega$ b) $4.3 \times 10^{-3} \Omega$ c) $1 \times 10^{-2} \Omega$ d) 4.44Ω
2. The current in a conductor varies with time t as $I = 2t + 3t^2$, where I is in ampere and t is in second. Electric charge flowing through a section of the conductor during $t = 2 \text{ sec}$ to $t = 3 \text{ sec}$ is _____.
a) 10 C b) 24 C c) 33 C d) 44 C
3. A short conductor of length 5.0 cm is placed parallel to a long conductor of length 1.5 cm near its centre. The conductors carry currents 4.0 A and 3.0 A respectively in the same direction. What is the total force experienced by the short conductor, when they are 3.0 cm apart? (take long wire as infinitely long)
a) $4 \times 10^{-8} \text{ N}$ b) $5 \times 10^{-8} \text{ N}$ c) $4.44 \times 10^{-8} \text{ N}$ d) $4 \times 10^{-6} \text{ N}$
4. A current of 10 A is passing through a long wire which has semi-circular loop of the radius 20 cm as shown in the figure. Magnetic field produced at the centre of the loop is _____.
a) $10 \pi \mu \text{ T}$ b) $5 \pi \mu \text{ T}$
c) $4 \pi \mu \text{ T}$ d) $2 \pi \mu \text{ T}$

5. Magnetic permeability is maximum for _____.
a) ferromagnetic substances b) diamagnetic substance
c) paramagnetic substances d) all of the above
6. If E and H represents the intensity of electric field and magnetising field respectively, then the unit of E/H will be :
a) ohm b) mho c) joule d) newton

Directions: The question no. 7 and 8 consist of two statements, Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (c) If the Assertion is correct but Reason is incorrect.
- (d) If both the Assertion and Reason are incorrect.

7. **Assertion:** The resistance of a conductor increases with temperature.

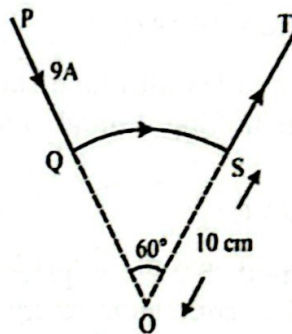
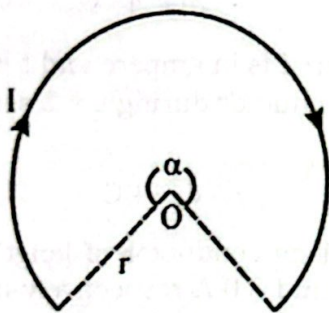
Reason: As temperature increases, the atoms in the conductor vibrate more, causing more collisions for electrons.

8. **Assertion :** The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason : The magnetic field inside the solenoid is uniform.

9. Explain about the limitations of ohm's law. (any two)

10. A circular segment of radius 10 cm subtends an angle of 60° at its centre. A current of 9 A is flowing through it. Find the magnitude and direction of the magnetic field produced at the centre



11.(i) Write the expression for the drift velocity of charge carriers in a conductor and explain how does drift velocity of electrons in a metallic conductor vary with increase in temperature?

(ii) A particle of charge $2 \mu\text{C}$ and mass 1.6 g is moving with a velocity $4 \hat{i} \text{ ms}^{-1}$. At $t = 0$ the particles enter in a region having an electric field $E = 80 \hat{i} + 60 \hat{j}$ (in N C^{-1}). Find the velocity of particle at $t = 5\text{s}$.

12. Apply Biot -savart law to find the magnetic field due to a circular current carrying loop at a point on the axis of the loop which is at a distance r from the centre.

13. Compare the properties - susceptibility ,permeability and effect of magnet of dia, para and ferro magnetic substances.

14. Read the following passage and write the answer for the questions given below from your understanding:

Current loop behaves like a magnetic dipole and has a magnetic field. They behave just like a magnet. Interesting part is, it depends upon the direction of current in a loop which decides whether the magnetic field line is in outward or inward direction. With the help of this outward and inward direction of magnetic field, North and South poles get decided. Anticlockwise direction of current creates north pole (outward

direction magnetic field) and clockwise direction of current creates a south pole (inward direction magnetic field). Magnetic dipole moment M with the circular current loop carrying a current I and of area A . The magnitude of m is given by $|m| = I \times A$. Current in the circular coil produces a magnetic field and amperes found that the magnetic field created due to the circular coil is similar to the magnetic field due to a bar magnet.

- i. A thin circular wire carrying a current I , has a magnetic moment M . The shape of a wire is changed to a square and it carries the same current. It will have a magnetic moment.

(a) $4M/\pi^2$ (b) M (c) $\pi M / 4$ (d) $4M/\pi A$

- ii. The unit of magnetic dipole moment is _____.

(a) ampere-square meter (Am^2) (b) joule tesla (JT)
(c) N/m (d) TmA^{-1}

- iii. A magnet of magnetic moment 2 J T^{-1} is aligned in the direction of magnetic field of 0.1 T . What is the net work done to bring the magnet normal to the magnetic field?

(a) 0.1 J (b) 0.2 J (c) 1 J (d) 2 J

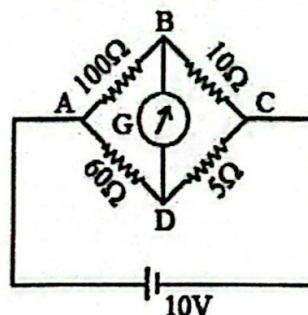
- iv. A bar magnet of magnetic moment m is placed in uniform magnetic field B such that m is parallel to B . In this position, the torque and force acting on it are _____ and _____ respectively.

(a) $0, 0$ (b) $m \times B, mB$ (c) $m \cdot B, mB$ (d) $m \times B, m \cdot B$

- 15.(i) State Kirchoff's current law.

- (ii) Apply Kirchoff's laws to derive the balance condition of wheatstone bridge.

- (iii) The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD . Calculate the current through the galvanometer when a potential difference of 10 V is maintained across AC .



- 16.(i) Describe the principle and working of a moving coil galvanometer with a neat diagram.

- (ii) Why is it not advisable to use galvanometer directly to measure current?

- (iii) If a current of $200 \mu\text{A}$ deflects the coil of a moving coil galvanometer through 60° , then the current required to cause deflection through $\pi/10$ radians is? What is the sensitivity of the galvanometer?

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CHENNAI - 600 063
I TERM EXAMINATION (2026 - 2027)
PHYSICS

ROLL NO. :

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CLASS: XII
DATE: 23.09.2026

MAXIMUM MARKS: 70
DURATION: 3 Hours

GENERAL INSTRUCTIONS:

- 1) There are 33 questions in all. All questions are compulsory.
- 2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- 3) All the sections are compulsory.
- 4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning questions of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study based questions of four marks each and Section E contains three long answer questions of five marks each.
- 5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- 6) Use of calculators is not allowed.
- 7) You may use the following values of physical constants wherever necessary

$$c = 3 \times 10^8 \text{ m/s}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$m_p = 1.7 \times 10^{-27} \text{ kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

SECTION-A

1. Two charges q_1 and q_2 are separated by a distance 'd' and placed in a medium of dielectric constant κ . What will be the equivalent distance between charges in air for the same electrostatic force?
A) $d\sqrt{\kappa}$ B) $\kappa\sqrt{d}$ C) $1.5 d\sqrt{\kappa}$ D) $2d\sqrt{\kappa}$
2. An electron experiences a force $(1.6 \times 10^{-16} \text{ N})\hat{i}$ in an electric field $\vec{E}$. The electric field $\vec{E}$ is:
A) $(1.0 \times 10^3 \frac{\text{N}}{\text{C}})\hat{i}$ B) $-(1.0 \times 10^3 \frac{\text{N}}{\text{C}})\hat{i}$ C) $(1.0 \times 10^{-3} \frac{\text{N}}{\text{C}})\hat{i}$ D) $-(1.0 \times 10^{-3} \frac{\text{N}}{\text{C}})\hat{i}$
3. Three capacitors, each of $4 \mu\text{F}$ are to be connected in such a way that the effective capacitance of the combination is $6 \mu\text{F}$. This can be achieved by connecting
A) All three in parallel.
B) All three in series.
C) Two of them connected in series and the combination in parallel to the third.
D) Two of them connected in parallel and the combination in series to the third.
4. Three point charges, each of charge q are placed on vertices of a triangle ABC, with $AB=AC=5L$, $BC=6L$. The electrostatic potential at midpoint of side BC will be
A) $\frac{11q}{48\pi\epsilon_0 L}$ B) $\frac{8q}{36\pi\epsilon_0 L}$ C) $\frac{5q}{24\pi\epsilon_0 L}$ D) $\frac{1q}{16\pi\epsilon_0 L}$

5. Which of the following statements is **not** true for electric energy in ac form compared to that in dc form?

- A) Production of ac is economical.
- B) ac can be easily and efficiently converted from one voltage to the other.
- C) ac can be transmitted economically over long distances.
- D) ac is less dangerous.

6. The rms and the average value of an ac voltage $V = V_0 \sin \omega t$ volt over a cycle respectively will be:

- A) $\frac{V_0}{2}, \frac{V_0}{\sqrt{2}}$
- B) $\frac{V_0}{\pi}, \frac{V_0}{2}$
- C) $\frac{V_0}{\sqrt{2}}, 0$
- D) $V_0, \frac{V_0}{\sqrt{2}}$

7. A plane electromagnetic wave travels through a medium and the magnetic field associated with it is given by $B = 5 \times 10^{-8} \sin(3 \times 10^{10} t - 150 x) T$. Where x is in metres and t is in seconds. The velocity of the wave is

- A) $2.0 \times 10^8 \text{ ms}^{-1}$
- B) $4.5 \times 10^7 \text{ ms}^{-1}$
- C) $3.5 \times 10^7 \text{ ms}^{-1}$
- D) $2.5 \times 10^8 \text{ ms}^{-1}$

8. Which of the following materials has positive and small value of magnetic susceptibility?

- A) Cu
- B) Al
- C) Bi
- D) Ni

9. A current carrying circular loop of magnetic moment $\vec{M}$ is suspended in a vertical plane in an external magnetic field $\vec{B}$ such that its plane is normal to $\vec{B}$. The workdone in rotating this loop by 45° about an axis perpendicular to $\vec{B}$ is closest to

- A) -0.3 MB
- B) 0.3 MB
- C) -1.7 MB
- D) 1.7 MB

10. What is the phase difference between electric ($\vec{E}$) and magnetic ($\vec{B}$) fields in an electromagnetic wave?

- A) Zero
- B) $\pi/2$
- C) π
- D) 2π

11. Rays which appear to converge or diverge from a focal plane travelling parallel to principal axis is called

- A) Convergent
- B) Paraxial
- C) Coaxial
- D) Divergent

12. A galvanometer of resistance 50Ω is converted into a voltmeter of range $(0 - 2 \text{ V})$ using resistor of $1.0 \text{ k}\Omega$. If it is to be converted into a voltmeter of range $(0 - 10 \text{ V})$, the resistance required will be

- A) $4.8 \text{ k}\Omega$
- B) $5.0 \text{ k}\Omega$
- C) $5.2 \text{ k}\Omega$
- D) $5.4 \text{ k}\Omega$

For Questions number 13 to 16, two statements are given one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

- A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- C) Assertion (A) is true, but Reason (R) is false.
- D) Both Assertion (A) and Reason (R) are false.

Assertion (A) : The current density ($\vec{j}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

Assertion (A) : The magnetic field produced by a current carrying solenoid is independent of its length and cross sectional area.

Reason (R) : The magnetic field inside the solenoid is uniform.

Assertion (A) : The mutual inductance between two coils is maximum when the coils are wound on each other.

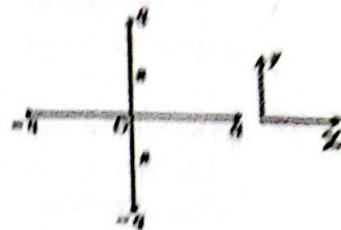
Reason (R) : The flux linkage between two coils is maximum when they are wound on each other

Assertion (A) : When a ferromagnetic substance is heated to high temperature it becomes paramagnetic in nature.

Reason (R) : The disappearance of magnetisation of a ferromagnet is abrupt and not gradual.

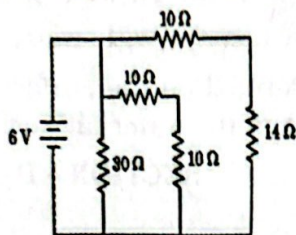
SECTION - B

Two identical dipoles are arranged in x-y plane as shown in the figure. Find the magnitude and the direction of net electric field at the origin O.



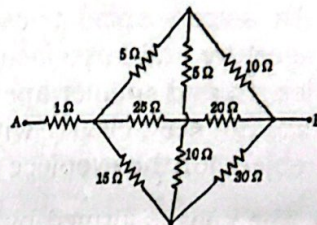
Name the electromagnetic wave used (i) in radar, (ii) as diagnostic tool in medicine. Write their wavelength range also.

Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.



OR

b) Find the equivalent resistance between points A and B for the network shown in the figure.



20. State Lenz's law. Give one example to illustrate this law.

The "Lenz's law is a consequence of the law of conservation of energy". Justify this statement.

21. Two thin lenses of focal length f_1 and f_2 are placed in contact with each other coaxially. Prove that the focal length f of the combination is given by $f = \frac{f_1 f_2}{f_1 + f_2}$

SECTION - C

22. State Gauss's law in electrostatics. Apply this to obtain the electric field $\vec{E}$ at a point near a uniformly charged infinite plane sheet.

23. Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (I) maximum, and (II) minimum?

24. a) (i) Write Biot-Savart law in the vector form. |

(ii) Two identical circular coils A and B, each of radius R , carrying currents I and $\sqrt{3}I$ respectively, are placed concentrically in XY and YZ planes respectively. Find the magnitude and direction of the net magnetic field at their common centre.

OR

b) An electron is moving in a circle with a speed $3.2 \times 10^7 \text{ ms}^{-1}$ in a magnetic field of $4.5 \times 10^{-4} \text{ T}$ perpendicular to it. Calculate:

- the radius of the circular path
- its frequency of rotation and
- its energy.

25. Define the term self-inductance and give its SI unit. Derive an expression for self-inductance of a long air cored solenoid of length l , cross sectional area A and having N number of turns.
26. a) Discuss the behaviour of an inductor connected to
(i) a dc source, and (ii) a high frequency ac source.
b) What is the phase relation between current and voltage in an ideal inductor connected to an ac source? Draw a phasor diagram for the circuit.
27. a) Depict the variation of electric field ($\vec{E}$) and magnetic field ($\vec{B}$) with respect to the direction of propagation of an electromagnetic wave. Write their two important characteristics.
b) Show that $\frac{1}{\sqrt{\epsilon_0 \mu_0}}$ gives the velocity of an electromagnetic wave in free space.
28. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface.

SECTION - D

CASE BASED QUESTIONS:

29. An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

The images formed by the objective lens and the eyepiece are respectively

- A) virtual, real B) real, virtual C) virtual, virtual D) real, real

The magnification produced by the telescope does **not** depend upon the

- A) colour of light B) focal length of objective lens
C) focal length of eyepiece D) apertures of objective lens and eyepiece

Which of the following statements is **not** correct for this telescope?

- A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).
B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).
C) The distance between two lenses is more than ($f_o + f_e$).
D) The magnifying power can be decreased by increasing the focal length of eyepiece

- (iv) a) An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of

- A) 84 cm B) 76 cm C) 20 cm D) 320 cm

OR

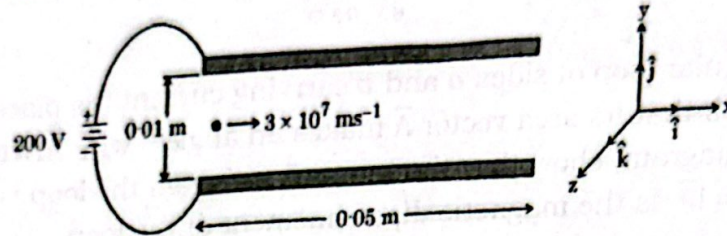
b) Consider the telescope described in question (iv) (a). Its magnifying power in normal adjustment will be

- A) 320 B) 84 C) 76 D) 20

30. The electric potential (V) and electric field (E) are closely related concepts in electrostatics. The electric field is a vector quantity that represents the force per unit charge at a given point in space, whereas electric potential is a scalar quantity that represents the potential energy per unit charge at a given point in space. Electric field and electric potential are related by the equations $E_r = \frac{-dV}{dr}$ and $\vec{E} = E_r \hat{r}$ i.e., electric field is the negative gradient of the electric potential. This means that electric field points in the

direction of decreasing potential and its magnitude is the rate of change of potential with distance. The electric field is the force that drives a unit charge to move from higher potential region to lower potential region and electric potential difference between the two points determines the work done in moving a unit charge from one point to the other point.

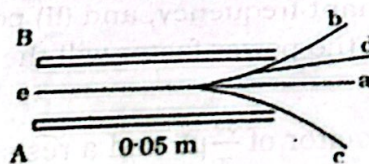
A pair of square conducting plates having sides of length 0.05 m are arranged parallel to each other in x-y plane. They are 0.01 m apart along z-axis and are connected to a 200 V power supply as shown in the figure. An electron enters with a speed of $3 \times 10^7 \text{ ms}^{-1}$ horizontally and symmetrically in the space between the two plates. Neglect the effect of gravity on the electron.



- (i) The electric field $\vec{E}$ in the region between the plates is
 A) $(2 \times 10^2 \frac{\text{V}}{\text{m}}) \hat{k}$ B) $-(2 \times 10^2 \frac{\text{V}}{\text{m}}) \hat{k}$ C) $(2 \times 10^4 \frac{\text{V}}{\text{m}}) \hat{k}$ ☒ D) $-(2 \times 10^4 \frac{\text{V}}{\text{m}}) \hat{k}$
- (ii) In the region between the plates, the electron moves with an acceleration $\vec{a}$
 A) $-3.5 \times 10^{15} \text{ ms}^{-2} \hat{k}$ ☒ B) $3.5 \times 10^{15} \text{ ms}^{-2} \hat{k}$
 C) $3.5 \times 10^{13} \text{ ms}^{-2} \hat{i}$ D) $-3.5 \times 10^{13} \text{ ms}^{-2} \hat{i}$
- (iii) a) Time interval during which an electron moves through the region between the plates is :
 A) $9.0 \times 10^{-9} \text{ s}$ B) $1.67 \times 10^{-8} \text{ s}$ ☒ C) $1.67 \times 10^{-9} \text{ s}$ D) $2.17 \times 10^{-9} \text{ s}$

OR

- b) The vertical displacement of the electron which travels through the region between the plates is :
 A) 10 mm ☒ B) 4.9 mm C) 5.9 mm D) 3.0 mm
- iv) Which one of the following is the path traced by the electron in between the two plates?



- ☒ A) a ☒ B) b C) c D) d

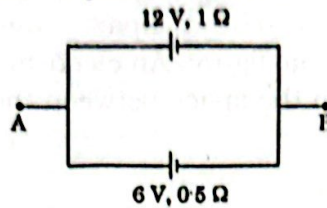
SECTION E

31. (a) (i) Consider a cylindrical conductor of length l and area of cross-section A . Current is maintained in the conductor and electrons drift with velocity v_d ($|v_d| = \frac{eE}{m} \tau$, where symbols have their usual meanings). Show that the conductivity σ of the material of the conductor is given by $\sigma = \frac{ne^2}{m} \tau$.
- (ii) The resistance of a metal wire at 20°C is 1.05Ω and at 100°C is 1.38Ω . Determine the temperature coefficient of resistivity of this metal.

OR

(b) (i) State Kirchhoff's current law and voltage law. Derive the condition for which the Wheatstone bridge is balanced.

(ii) Two cells of emfs 12 V and 6 V are connected in parallel as shown in the figure. Their internal resistances are $1\ \Omega$ and $0.5\ \Omega$ respectively. Calculate the emf and internal resistance of the equivalent cell between points A and B.



32.(a) (i) A rectangular loop of sides a and b carrying current I is placed in a magnetic field $\vec{B}$ such that its area vector $\vec{A}$ makes an angle θ with $\vec{B}$. With the help of a suitable diagram, show that the torque $\vec{\tau}$ acting on the loop is given by $\vec{\tau} = \vec{m} \times \vec{B}$ where $\vec{m} = I\vec{A}$ is the magnetic dipole moment of the loop.

(ii) A circular coil of 100 turns and radius $\frac{10}{\sqrt{\pi}}$ cm carrying current of 5.0 A is suspended vertically in a uniform horizontal magnetic field of 2.0 T. The field makes an angle 30° with the normal to the coil. Calculate

(I) the magnetic dipole moment of the coil, and

(II) the magnitude of the counter torque that must be applied to prevent the coil from turning.

OR

(b) (i) Derive an expression for the force $\vec{F}$ acting on a conductor of length L and area of cross-section A carrying current I and placed in a magnetic field $\vec{B}$.

(ii) What is the magnitude of magnetic force per unit length on a wire carrying a current of 8 A and making an angle of 30° with the direction of a uniform magnetic field of 0.15 T?

33. (a) (i) Define the terms (I) resonant frequency, and (II) power factor of a series LCR circuit. For what value of the power factor will the power dissipated in the circuit be maximum.

(ii) An inductor of $\frac{5}{\pi}$ H, a capacitor of $\frac{50}{\pi}\ \mu\text{F}$ and a resistor of $400\ \Omega$ are connected in series across an ac voltage $v = 140 \sin(100\pi t)$ V. Calculate :

(I) impedance of the circuit, and (II) rms value of current that flows in the circuit.

(Take $\sqrt{2} = 1.4$)

OR

(b) (i) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary voltage to primary voltage in terms of number of turns in the two coils.

(ii) The number of turns in the primary and the secondary coil of an ideal transformer are 100 and 5000 respectively. If 3.3 kW power is supplied to the transformer at 220 V, find (I) current in the primary coil, and (II) output voltage
